

Neutron Transport and Tritium Breeding Modelling in Nuclear Fusion Reactor Breeder Blankets

MPhys Project Report

Vadan Khan (10823198)

In collaboration with Rhodri Peters

4th Year MPhys, Department of Physics and Astronomy, University of Manchester

April 13, 2026

Abstract

meow

CONTENTS

1	Introduction	1
2	Theory	1
2.1	Nuclear Physics	1
2.2	ML: Bayesian Optimisation	1
3	Literature Review	1
3.1	Breeder Blankets	1
3.2	Surrogate Modelling in Science	1
3.3	ML Optimisation in Science	1
4	Methodology	1
4.1	Neutronic Analysis	1
4.2	ML Optimisation	1
5	Results	1
5.1	Layered Blanket Model Neutronics	1
5.2	Manual Thickness Distribution Analysis	2
5.3	ML Optimisation Findings	2
6	Discussion	2
7	Conclusion	2
	References	2

1. INTRODUCTION

Start your new document here! Moir, 1982, Momoshima (2022), (Lee, 1983), (Segantin et al., 2020)

2. THEORY

2.1. Nuclear Physics

2.2. ML: Bayesian Optimisation

3. LITERATURE REVIEW

3.1. Breeder Blankets

3.2. Surrogate Modelling in Science

3.3. ML Optimisation in Science

4. METHODOLOGY

4.1. Neutronic Analysis

4.2. ML Optimisation

Explaining what we explored for ML Explaining how we scaled up and tested higher dimensions

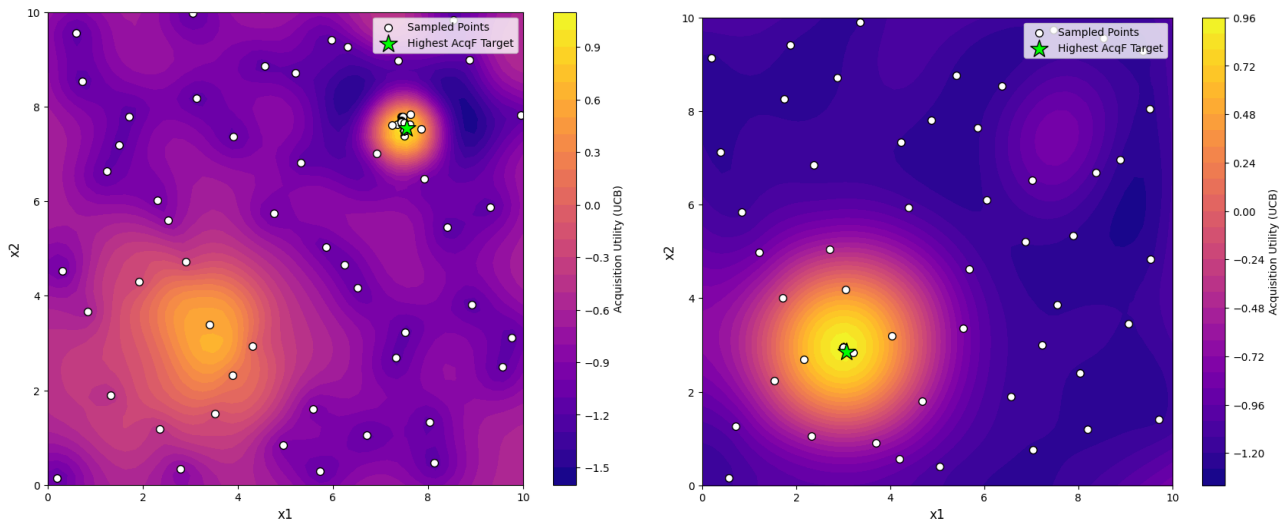


Figure 1: Lets write a dummy caption here just to test and view how a caption would look and how much impact a long caption would have. It is important when we are testing to account for the impact that these could have as it would lead to bad planning when it comes to spacing.

and functions.

5. RESULTS

5.1. Layered Blanket Model Neutronics

result intro highlighting key results highlighting key results highlighting key results

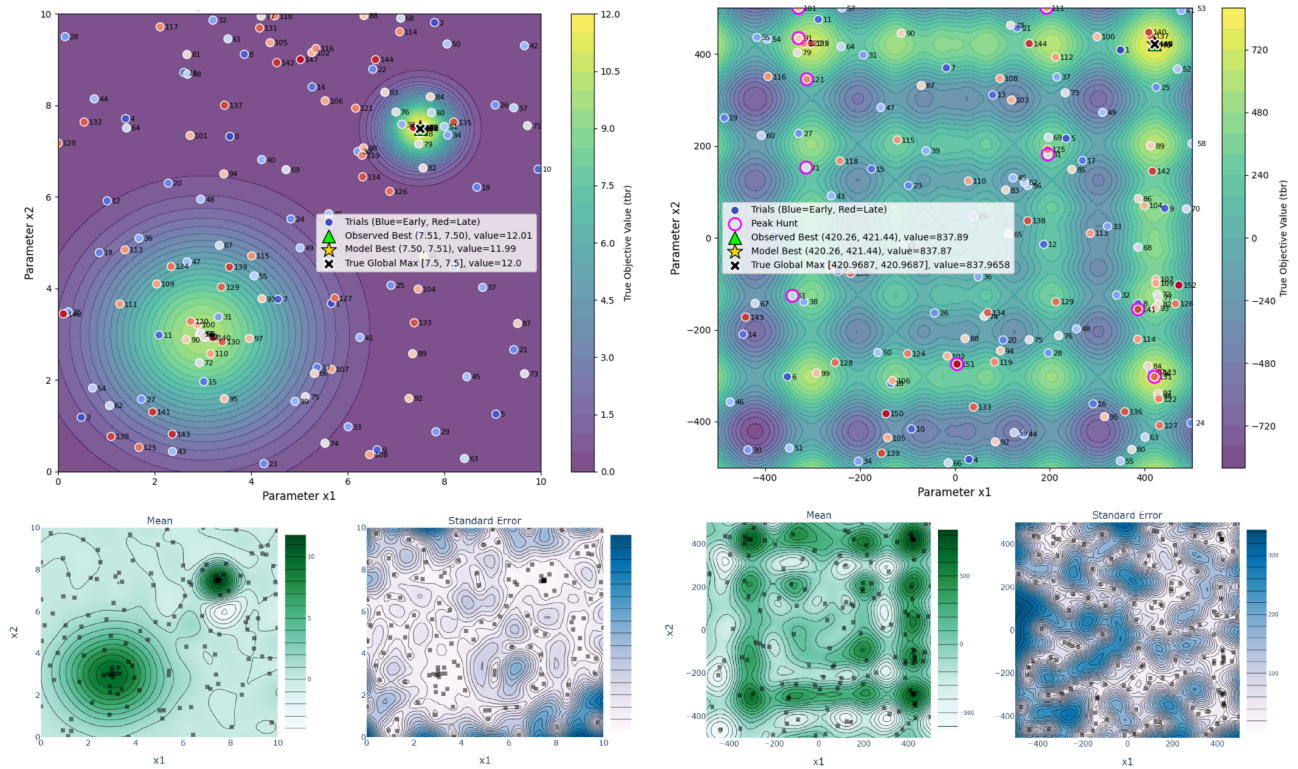


Figure 2: Lets write a dummy caption here just to test and view how a caption would look and how much impact a long caption would have. It is important when we are testing to account for the impact that these could have as it would lead to bad planning when it comes to spacing.

5.2. Manual Thickness Distribution Analysis

highlighting key results highlighting key results highlighting key results highlighting key results

5.3. ML Optimisation Findings

highlighting key results highlighting key results highlighting key results

6. DISCUSSION

7. CONCLUSION

REFERENCES

Note: DOI links are hyperlinked within “[DOI]” placeholder

- Lee, J. D. (1983). ‘U.S.-DOE Fusion-Breeder Program: Blanket Design and System Performance’, *Proceedings of the Third International Conference on Emerging Nuclear Energy Systems*, pp. 1–37. Available at: <https://www.osti.gov/servlets/purl/6012859>.
- Moir, R. W. (1982). ‘The fusion breeder’, *Journal of Fusion Energy*, 2(4), pp. 351–367. [DOI].
- Momoshima, N. (2022). ‘TRITIUM IN THE ENVIRONMENT’, *Radiation Protection Dosimetry*, 198(13), pp. 896–903. [DOI].
- Segantin, S. et al. (2020). ‘Optimization of tritium breeding ratio in ARC reactor’, *Fusion Engineering and Design*, 154, pp. 1–5. [DOI].

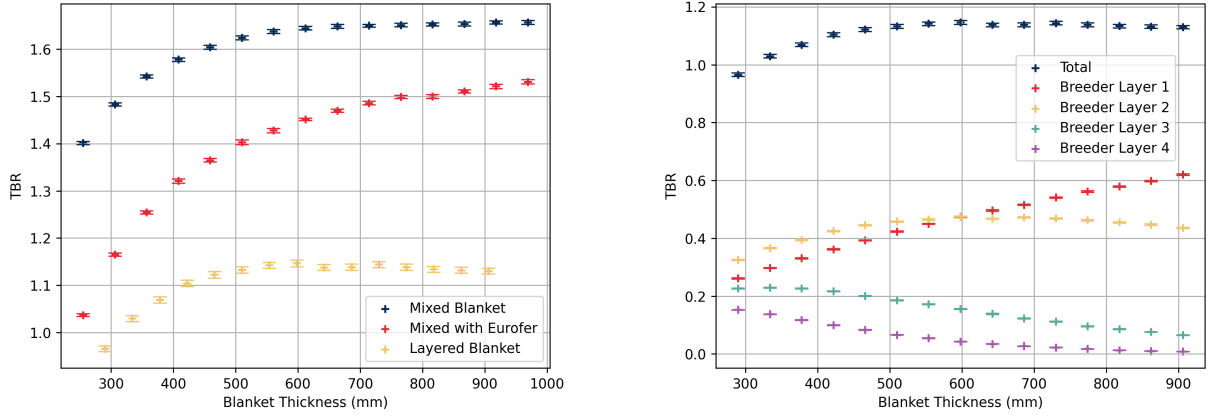


Figure 3: Lets write a dummy caption here just to test and view how a caption would look and how much impact a long caption would have. It is important when we are testing to account for the impact that these could have as it would lead to bad planning when it comes to spacing.

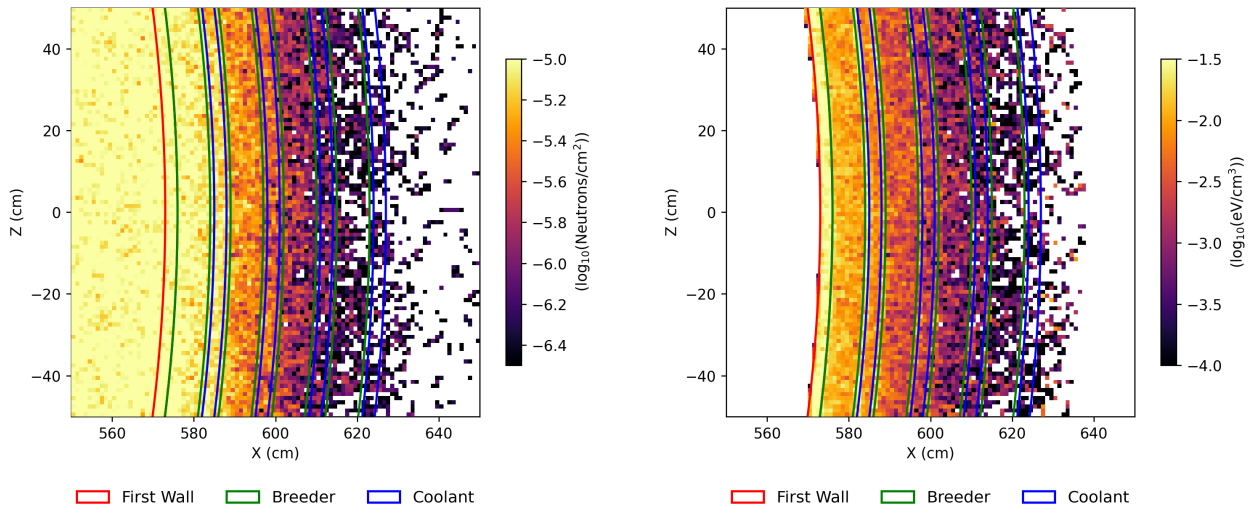


Figure 4: Lets write a dummy caption here just to test and view how a caption would look and how much impact a long caption would have. It is important when we are testing to account for the impact that these could have as it would lead to bad planning when it comes to spacing.

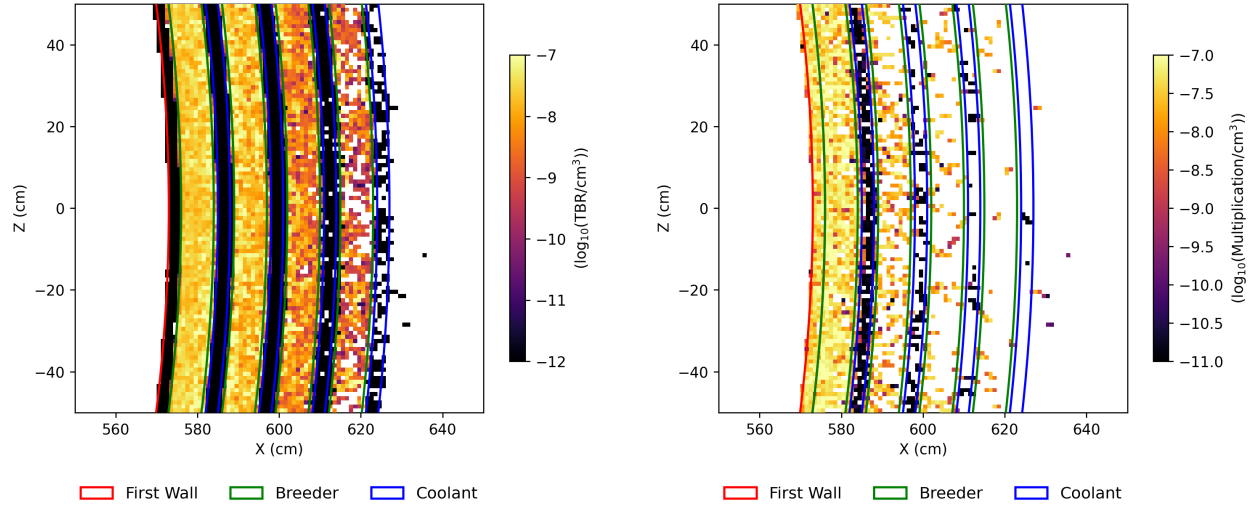


Figure 5: Lets write a dummy caption here just to test and view how a caption would look and how much impact a long caption would have. It is important when we are testing to account for the impact that these could have as it would lead to bad planning when it comes to spacing.

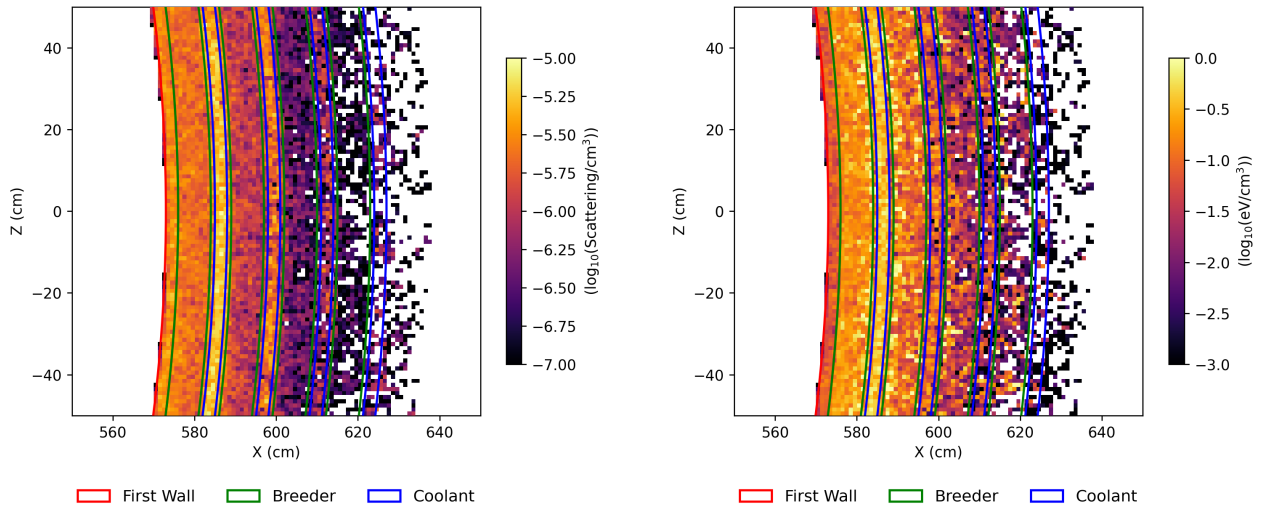


Figure 6: Lets write a dummy caption here just to test and view how a caption would look and how much impact a long caption would have. It is important when we are testing to account for the impact that these could have as it would lead to bad planning when it comes to spacing.

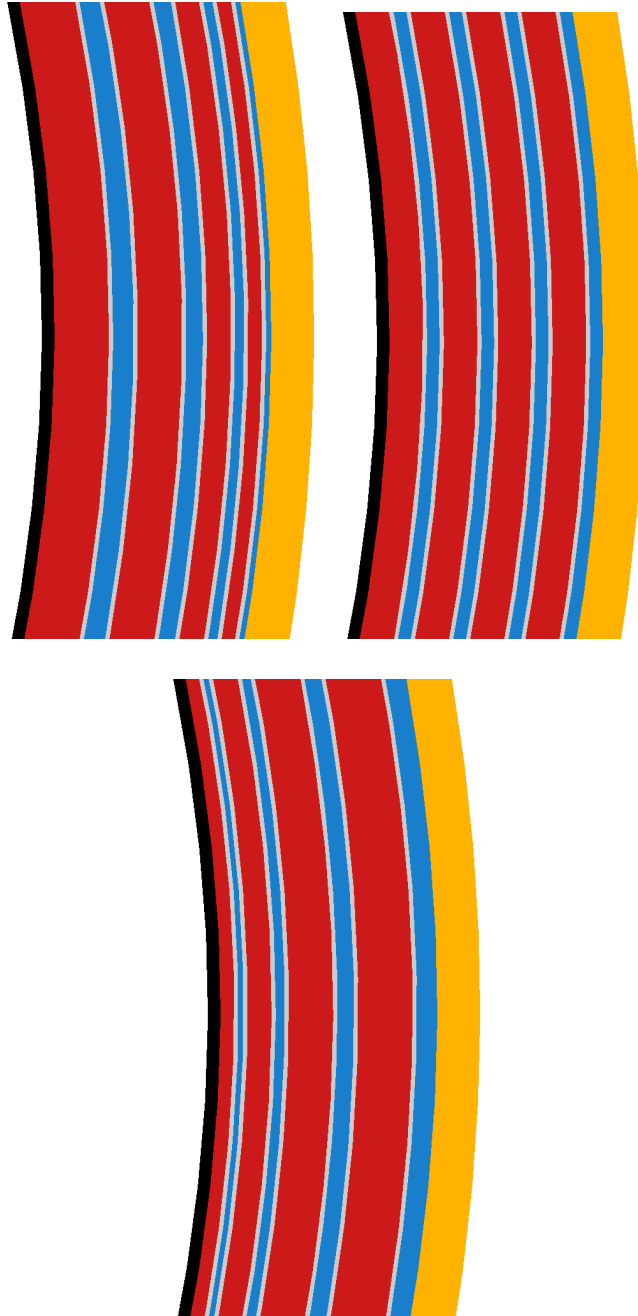


Figure 7: Lets write a dummy caption here just to test and view how a caption would look and how much impact a long caption would have. It is important when we are testing to account for the impact that these could have as it would lead to bad planning when it comes to spacing.

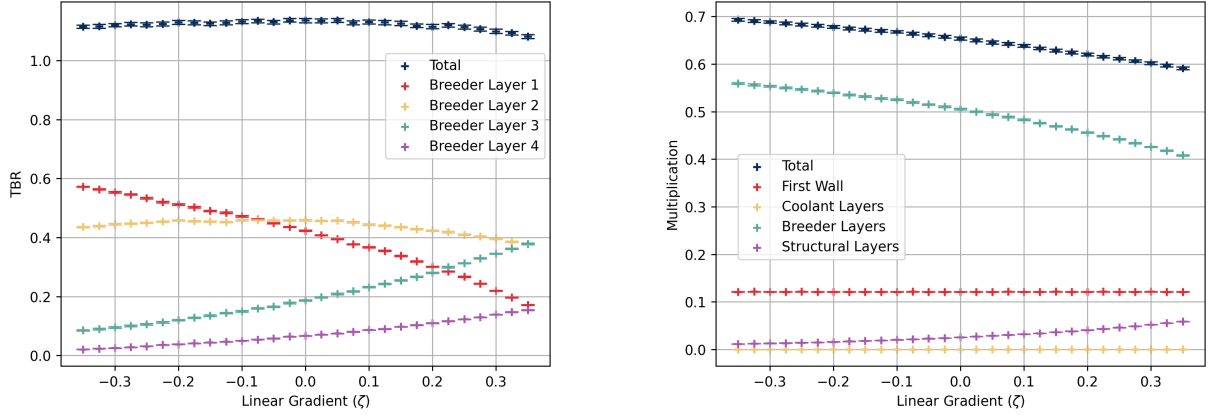


Figure 8: Lets write a dummy caption here just to test and view how a caption would look and how much impact a long caption would have. It is important when we are testing to account for the impact that these could have as it would lead to bad planning when it comes to spacing.

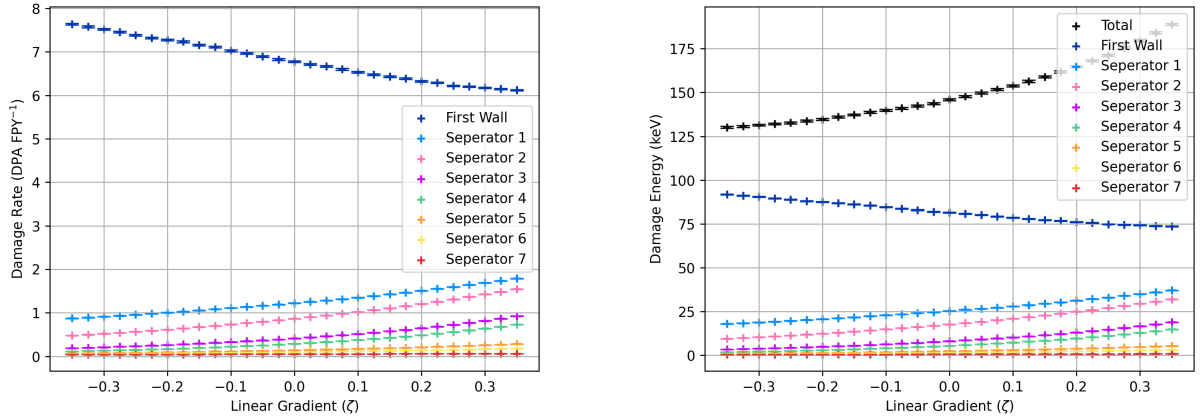


Figure 9: Lets write a dummy caption here just to test and view how a caption would look and how much impact a long caption would have. It is important when we are testing to account for the impact that these could have as it would lead to bad planning when it comes to spacing.

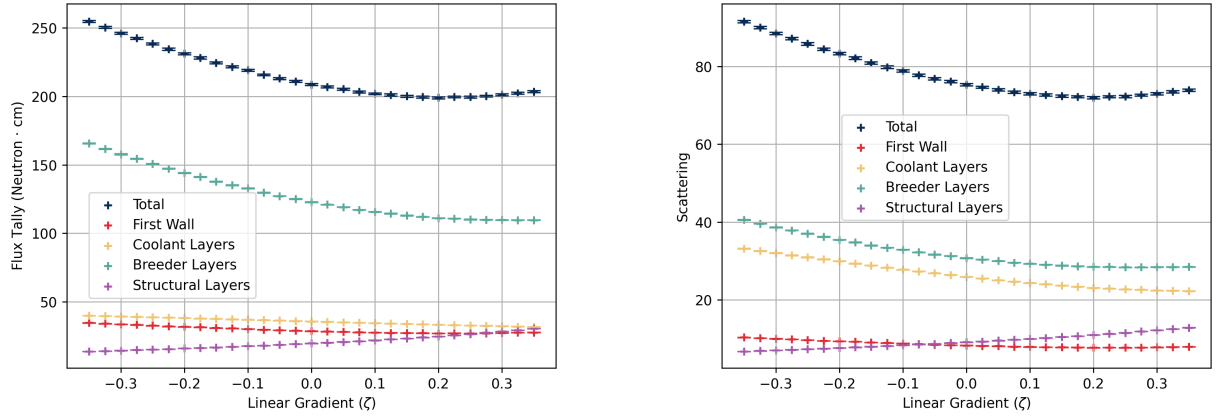


Figure 10: Lets write a dummy caption here just to test and view how a caption would look and how much impact a long caption would have. It is important when we are testing to account for the impact that these could have as it would lead to bad planning when it comes to spacing.

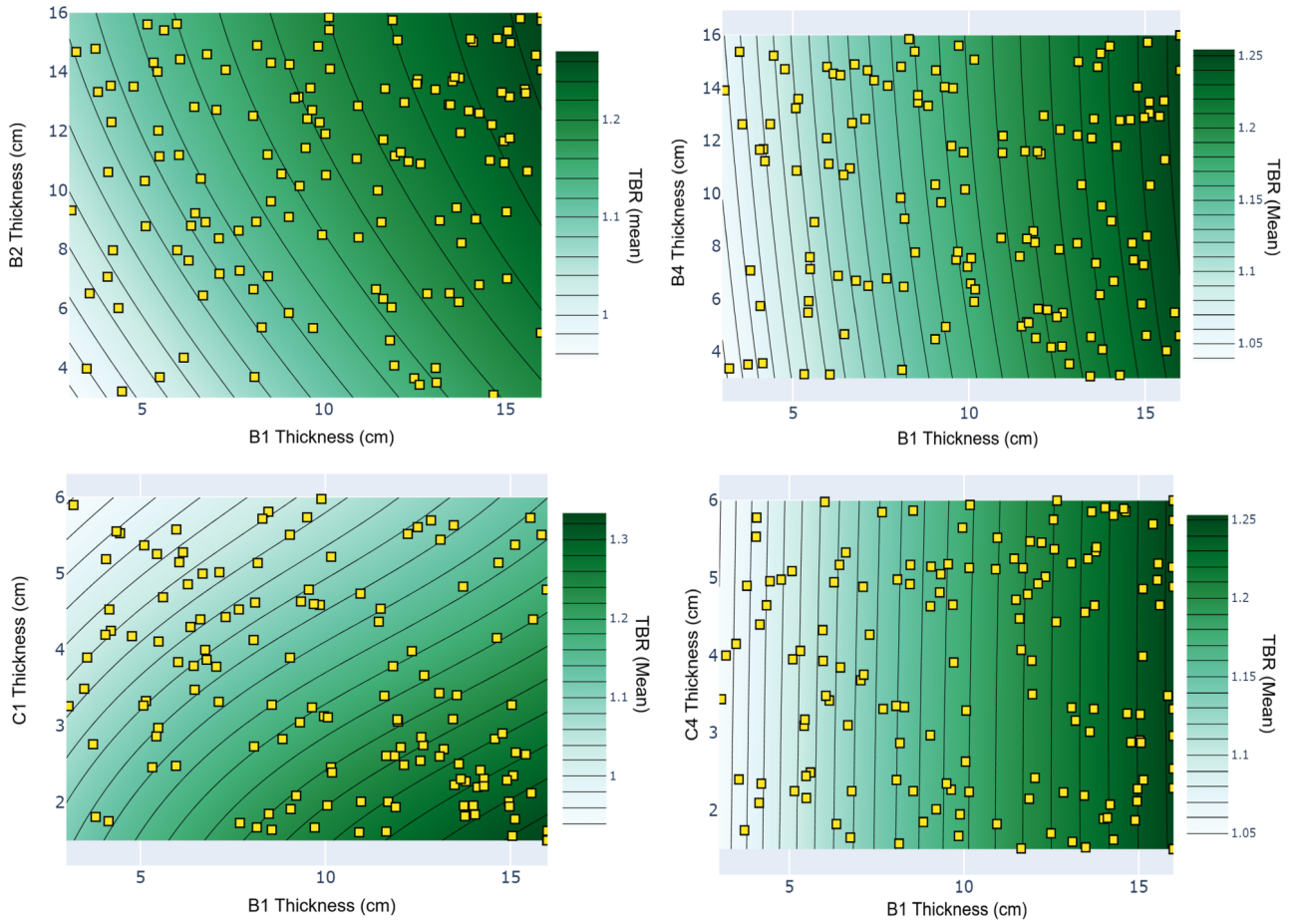


Figure 11: Lets write a dummy caption here just to test and view how a caption would look and how much impact a long caption would have. It is important when we are testing to account for the impact that these could have as it would lead to bad planning when it comes to spacing.

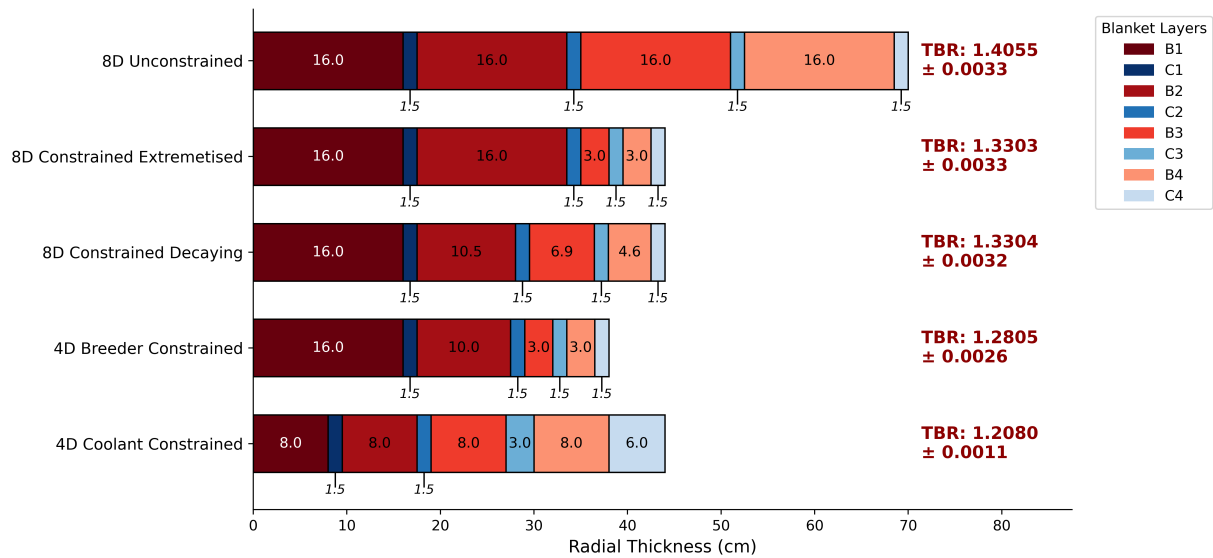


Figure 12: Lets write a dummy caption here just to test and view how a caption would look and how much impact a long caption would have. It is important when we are testing to account for the impact that these could have as it would lead to bad planning when it comes to spacing.